

Wi-Fi Security Protocol

Module 32

Duration: 26:30

Video: <https://www.youtube.com/watch?v=vCdk5ufJpPo>

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0:09 [Music]

0:24 n

0:34 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:42 University of Science and Technology Hisar harana today in this lecture we

0:49 will study about WiFi security protocols first of all please see the contents

0:56 which I shall cover after covering brief give introduction about three basic Wi-Fi security protocols I shall cover

1:05 each one by one first in this series will be wired equivalent privacy

1:15 we followed by WiFi protected access WPA and

1:24 WPA2 in the last I shall compare WPA and WPA2

1:32 technology the wireless networking is becoming very popular among large number

1:38 of Internet users because of the popularity of the wireless networking

1:45 technology large number of wireless networking products and Protocols are

1:52 available for the home and Business use wireless Transmissions are available

2:00 to authorized users and also to the unauthorized users like

2:08 hackers the IEEE 802.11 standard offers some level of

2:16 protection this protection known as the wired equivalent privacy WPA protocol

2:24 defines a set of instructions and rules by which wireless data can be

2:32 transmitted over the air waves with some amount of

2:38 security wired equivalent privacy WPA protocol is a basic security feature in

2:46 the IEEE 802.11 standard intended to provide

2:54 confidentiality over a wireless network by encrypting information sent over the

3:01 network very soon web got replaced by Wi-Fi protocol access

3:08 WPA due to a flaw in web of getting cracked in a few minutes with the aid of

3:16 automated tools WiFi protected access WPA and

3:24 WPA2 is a wireless security protocol designed to address

3:30 and fix the known security issues in we anyway this was very brief about
3:40 basic security techniques used for wifi now let us see each one by one in
3:48 detail first is wired equivalent privacy web which is a security algorithm for i
3:59 e a 802.11 wireless networks introduced as part of the original
4:07 802.11 standard ratified in 1997 its intention was to provide data
4:17 confidentiality comparable to that of a traditional wired Network the design
4:24 objectives of web are as per i e 82. 11
4:30 standard which states the following reasonably strong the security afforded
4:38 by the algorithm should rely on the difficulty of discovering the secret key
4:45 through a Brute Force attack this in turn is related to the length of the
4:51 secret key and the frequency of changing Keys self synchronizing web should be
self-
5:01 5 minutes, 1 second synchronizing for each message this property is critical for a
data link
5:08 level encryption algorithm where best effort delivery is
5:15 assumed efficient the web algorithm should be efficient and may be implemented in
5:24 either Hardware or software exportable
5:31 every effort should be made to design the web system operation so as to
5:38 maximize the chances of approval by the US government for exporting this
5:46 technology optional the implementation and use of web in I
5:55 802.11 should be optional from the above objectives it's clear
6:03 that VP was not designed to provide a high military level
6:09 security the intention was to make it hard to break in as opposed to
6:17 impossible to break in now let us move on to the technical details of
6:26 web web security involves two Parts
6:34 authentication and encryption authentication in web involves authenticating a
device when it
6:43 first joins the Lan the authentication process in the wireless network using web is
to prevent
6:52 devices or stations joining the network unless they know the web
7:01 7 minutes, 1 second key two methods of authentication can be used with web Open
System Authentication
7:09 and shared key authentication in Open System Authentication the W Lan client
need not
7:19 provide its credentials to the access point during authentication any client can
7:27 authenticate with the access point and then attempt to associate in fact no
7:36 authentication occurs subsequently VB keys can be used for encrypting data
7:45 frames at this point the client must have the correct

7:52 keys in shared key authentication the web key is used for authentication in a
7:59 for Four Step Challenge response handshake as shown in figure step one
8:08 wireless device sends authentication request to the wireless access point
8:15 step two in response wireless access point sends 128bit random authentication
8:23 Challenge in a clear text to the requesting Wireless
8:31 device step three the wireless device uses the shared secret key to sign the
8:39 challenge and send them in form of authentication response to the wireless
8:46 access point step four wireless access point decrypts the signed message using the
8:55 shared secret key and verifies the challenge that that it has sent
9:01 9 minutes, 1 second before if the challenge matches then authentication succeeds
otherwise not if
9:09 authentication succeeds then authentication success message is sent
9:15 from access point to wireless device in webbased security
9:23 unfortunately no secret key is exchanged after Authentication the same secret key
or shared key is
9:33 used for both authentication and encryption so there is no way to tell
9:42 whether the subsequent message comes from the trusted device or from an
9:50 imposter this kind of authentication is prone to man in the middle
9:57 attack next we shall take up encryption part of web please see figure which shows
10:06 stepbystep implementation of encryption process web uses rc4 stream Cipher to
10:16 encrypt data between access point and wireless
10:21 device web uses 8bit rc4 and operates on
10:27 8bit values by creating an array with 256 8 bit values for a lookup
10:36 table VB uses CRC for the data integrity and accordingly performs cyclic
10:44 redundancy check operation on the plain text and generates CRC
10:51 value the CRC value is concatenated to the plain text the secret key is Con
10:59 catenated to the initialization Vector i v and fed into the
11:07 rc4 based on the secret key and IV rc4 generates key
11:16 stream the key stream and plain Text Plus CRC message are exod together the
11:26 result is the cipher text the same initialization Vector that was
11:33 used before is prepended in clear text to the resultant Cipher
11:41 text the initialization Vector IV plus Cipher text along with the frame headers
11:49 are then transmitted over the air in this encryption method as we have
11:57 seen some part of the secret key is used with different exposed values and
12:05 attacker can generate the secret part by analyzing some portion of number of bits
12:13 in the first few bytes of the key stream with relatively less
12:19 work another problem with weap is that the secret shared key is concatenated

12:27 with the visible IV value this weakness is known as IV
12:35 weakness in the beginning it was believed that WEP offers impenetrable
12:42 resistance to the eavesdroppers and hackers however as wireless networks
12:49 began to grow in popularity many Cryptanalysts and researchers discovered flaws
in the
12:57 original WEP design many believe that there was little peer review performed on
the WEP
13:08 protocol many of the WEP flaws would have been caught in the early design
13:14 phase if its design and implementation specifications had been reviewed
13:22 thoroughly for most of the wireless networking users especially home users
13:29 WEP is the only choice available until new security mechanisms are added to the
13:37 IEEE 802.11 standard but as people say something is better than nothing even
13:46 with its known weakness WEP was still more effective than no security at all
13:55 at least it will provide some security against unauthorized use of wireless
14:01 networks 14 minutes, 1 second network and eating up the bandwidth there are many
solutions available to
14:09 overcome the weakness of WEP some of them are the bigger size of the
14:15 initialization vector IV can be chosen the hashed value of IV can be
14:23 prepended or appended to the cipher text instead of the clear
14:30 text instead of using CRC checksum different method can be used for the
14:37 data integrity verification that is Hash functions change secret key regularly
14:46 dynamically using secure symmetric key distribution
14:53 protocols better key management using security handshake
15:00 protocols new authentication mechanisms using the extensible Authentication
Protocol
15:09 EAP in spite of implementation of these improvements there was a flaw in WEP
15:15 of getting cracked in a few minutes with the aid of automated tools so instead of
15:23 just relying on the WEP security alone additional measures must be taken to
15:30 provide better security among wireless devices in 2003 the Wi-Fi Alliance
15:40 announced that WEP had been superseded by Wi-Fi protected access
15:48 WPA in 2004 with a ratification of the full
15:55 IEEE 802.11i standard that is WPA2 the IEEE declared that WEP has been
16:03 declared obsolete Wi-Fi Alliance also decided to implement IEEE
16:11 802.11i standard to overcome the problems in WEP
16:20 IEEE 802.11i was implemented in two steps Wi-Fi protected access WPA and Wi-Fi
16:29 protected access 2 that is WPA2 the Wi-Fi Alliance intended WPA as an
16:39 intermediate measure to take the place of WEP pending the availability of full

16:46 I E 802.11i standard WPA could be implemented
16:55 through firmware upgrades on wireless network interface cards designed for web
17:03 as WPA was an intermediate solution for Hardware that could not support WPA 2
so
17:11 to implement WPA an interim software implementable solution was designed that
17:21 could delay immediate deployment of new hardware to add support vot for WPA or
17:30 WPA 2 some old Wi-Fi access points might need to be replaced or have their
17:39 firmware upgraded WPA which is sometime referred to as draft i e
17:48 802.11 i standard became available in
17:54 2003 WPA2 became available in 20 2004
18:00 and is commonly known as full i e
18:06 802.11i or i e 802.11i 2004
18:14 standard WPA is a wireless security protocol designed to address and fix the
18:22 known security issues in web WPA provides users with a higher
18:30 level of assurance that their data will remain protected by using temporal key
18:38 Integrity protocol tkip for data encryption specifically the temporal key
18:47 Integrity protocol tkip was adopted for
18:53 WPA we used a 40 bit or 100 4bit
19:00 encryption key that must be manually entered on wireless access points and
19:07 devices which does not change but tkip employs a per packet key
19:16 meaning that it dynamically generates a new 128bit key for each packet and thus
19:25 prevents the type of attack that compromise we WPA also includes a message
Integrity
19:36 check which is designed to prevent an attacker from altering and resending data
19:44 packets this replaces the cyclic redundancy check CRC that was used by the web
standard
19:54 crc's main flaw in web was that it did not provide provide a sufficiently strong data
Integrity
20:03 guarantee for the packets it handled well tested message authentication codes
existed to solve
20:13 these problems but they required too much computation to be used on Old
20:20 network cables WPA uses a message Integrity
20:26 check algorithm called Michael to verify the Integrity of the
20:33 packets no doubt Michael is much stronger than a CRC but still
20:39 researchers have discovered a flaw in WPA due to reason of limitations of
20:47 Michael to retrieve the key stream from short packets to use for
20:54 reinjection and spoofing in order to solve such problems
21:01 21 minutes, 1 second in WPA and to completely Implement i e

21:07 802.11i protocol a new wireless security protocol
21:13 WPA2 is implemented in which only authorized users can access a wireless
21:22 device with features supporting stronger cryptography stronger Authentication
21:29 control key management replay attack protection and data
21:37 Integrity going into technical aspects of WPA 2 instead of
21:44 tkip advanced encryption standard AES is used for encryption purpose that has a
21:53 fixed block size of 128 bits and three different key sizes using three
22:03 different steps of the algorithm in the first iteration a
22:10 128bit key is used to perform 9 rounds in the second a
22:18 192bit key performs 11 rounds and in the third iteration a
22:27 256bit key is used to perform 133 rounds because AES is a substitution
22:36 Cipher So within each round bits are substituted and rearranged and then
22:45 special multiplication is performed based on the new Arrangement the
effectiveness of AES
22:54 cannot be disputed as the time needed to break it by using a Brute Force attack
23:02 with a 128bit key length is around 2200
23:09 years now after brief technical details let us compare WPA and WPA 2 there are
23:19 some similarities and differences between WPA and WPA 2
23:29 both the security techniques WPA and WPA 2 uses the
23:37 802.1x that is EAP framework as part of the infrastructure that ensures Central
23:46 Mutual authentication and dynamic Key Management both offers a pre-shared key
23:56 for using home and small office environments and both are designed to
24:03 secure all versions of 802.11 devices including
24:11 802.11b 802.11 A and 802.11
24:17 G now coming onto the differences in both techniques WPA2 uses a mix mode
24:27 that supports both WPA and WPA2 enabled
24:33 devices on the same wireless network but WPA supports only
24:40 one however the most significant difference between WPA and WPA2 is that
24:49 WPA2 uses the advanced encryption
24:53 standard AES instead of tki I for data
25:01 25 minutes, 1 second encryption WPA2 is theoretically not hackable but WPA
25:10 is WPA2 requires more processing power than WP
25:18 a so dear friends today in this lecture we have covered introductory concepts of
25:26 wi-fi security followed by by three basic security techniques VB WPA and WPA
25:36 2 used for Wi-Fi in last we have also seen comparison of two latest security
25:45 techniques WPA and WPA 2 so this was all
25:52 about WiFi security protocols thank you e

25:59

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